Multi-Model Machine Learning Identifies MAPT, CHEK1, AURKA as Breast Cancer Prognostic Markers

Computational Biology and Bioinformatics
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Abstract

Breast cancer is the most common cancer among women globally, and the second leading cause of cancer death. While early detection improves survival rates, personalizing treatment based on genomic biomarkers could further increase survival duration. This project applies a multi-model machine learning approach to identify key gene biomarkers correlating with breast cancer survival duration, enabling physicians to personalize treatment. I hypothesized that different models contribute complementary findings: linear regression captures proportional gene-survival relationships, random forest reveals non-linear interactions, and neural network could perform deep analysis on larger datasets but was anticipated to underperform on the small METABRIC dataset (n=1904 patients).

Using the METABRIC dataset, I preprocessed gene expression and clinical data by imputing missing values and encoding categorical variables. I trained three models—ElasticNet linear regression, random forest, and PyTorch neural network—using 70/15/15 train/validation/test split to predict overall survival time.

Random forest achieved the best test performance (r^2=0.147), followed by linear regression (r^2=0.138), while the neural network underperformed (r^2=0.052) as anticipated. The models identified MAPT, CHEK1, and AURKA as top gene biomarkers strongly associated with survival duration, consistent with published cancer genomics research.

The analysis focused on genomic factors, yet survival duration also depends on age, comorbidities, lifestyle, and unrelated causes of death, introducing variance. The consistent under-prediction of survival durations aligns with this limitation and validates the integrity of our genomic-focused approach. These findings demonstrate that complementary models can uncover actionable genomic biomarkers, offering a pathway toward personalized breast cancer treatment and improved prognosis outcomes.

Acknowledgements

This project used the publicly available METABRIC dataset (Kaggle) and open-source Python libraries including PyTorch, scikit-learn, pandas, NumPy, SciPy, matplotlib, and graphviz.

Introduction

Breast cancer is the most frequently diagnosed cancer among women worldwide, with approximately 310,720 new cases expected in the United States in 2024 alone (NCI SEER). It is the second leading cause of cancer death among women. While overall five-year survival rates reach 90% when detected early, outcomes vary dramatically based on cancer subtype, stage, and individual patient characteristics. This variation underscores the urgent need for precision medicine approaches that can predict how long a patient will survive and tailor treatment accordingly.

Review of Literature

Existing research in cancer genomics has demonstrated that gene expression profiles can reveal molecular subtypes of breast cancer with distinct clinical outcomes. The METABRIC consortium (Curtis et al., 2012; Pereira et al., 2016) characterized the genomic and transcriptomic landscape of over 2,000 breast tumors, identifying novel subgroups with different survival patterns. However, most machine learning applications in this field focus on predicting binary survival outcomes (alive vs. deceased) rather than continuous survival duration, which is far more clinically valuable for treatment planning. This project addresses that gap by using three complementary machine learning models to predict survival time and to discover which specific genes most strongly influence survival duration, enabling physicians to customize treatment plans for individual patients.

Statement of Problem

Can multi-model machine learning identify gene biomarkers that correlate with breast cancer survival duration to enable personalized treatment?

Hypothesis

I hypothesized that different models would contribute multi-dimensional findings. Linear regression identifies genes acting proportionally on survival duration, while random forest captures non-linear and interaction-based relationships for comprehensive coverage. A neural network could perform deep analysis on very large datasets to discover additional differential factors among large populations. However, because the METABRIC dataset contains fewer than 2,000 records, the neural network was anticipated to underperform relative to the other models due to insufficient data volume for deep learning.

Materials

Category	Component / Tool	Specification / Purpose
HARDWARE	Microsoft Surface Pro 9	Windows 11 laptop; used for coding, tool access, and results review.
	Google Colab	Cloud environment; Python T4 GPU for model training and evaluation.
DATASET	METABRIC	1,904 patients; clinical features and 489 gene expression data (Kaggle).
SOFTWARE	Python 3	Main programming language for data analysis and ML workflows.
	PyTorch (torch)	Deep learning framework for building and training neural networks.
	scikit-learn	Library for regression, random forest, tuning, and evaluation.
	pandas / NumPy	Data cleaning, organization, and numerical operations.
	SciPy	Statistical calculations including confidence intervals.
	matplotlib	Data visualization and graph generation.
	graphviz	Decision tree surrogate visualization.

Procedures

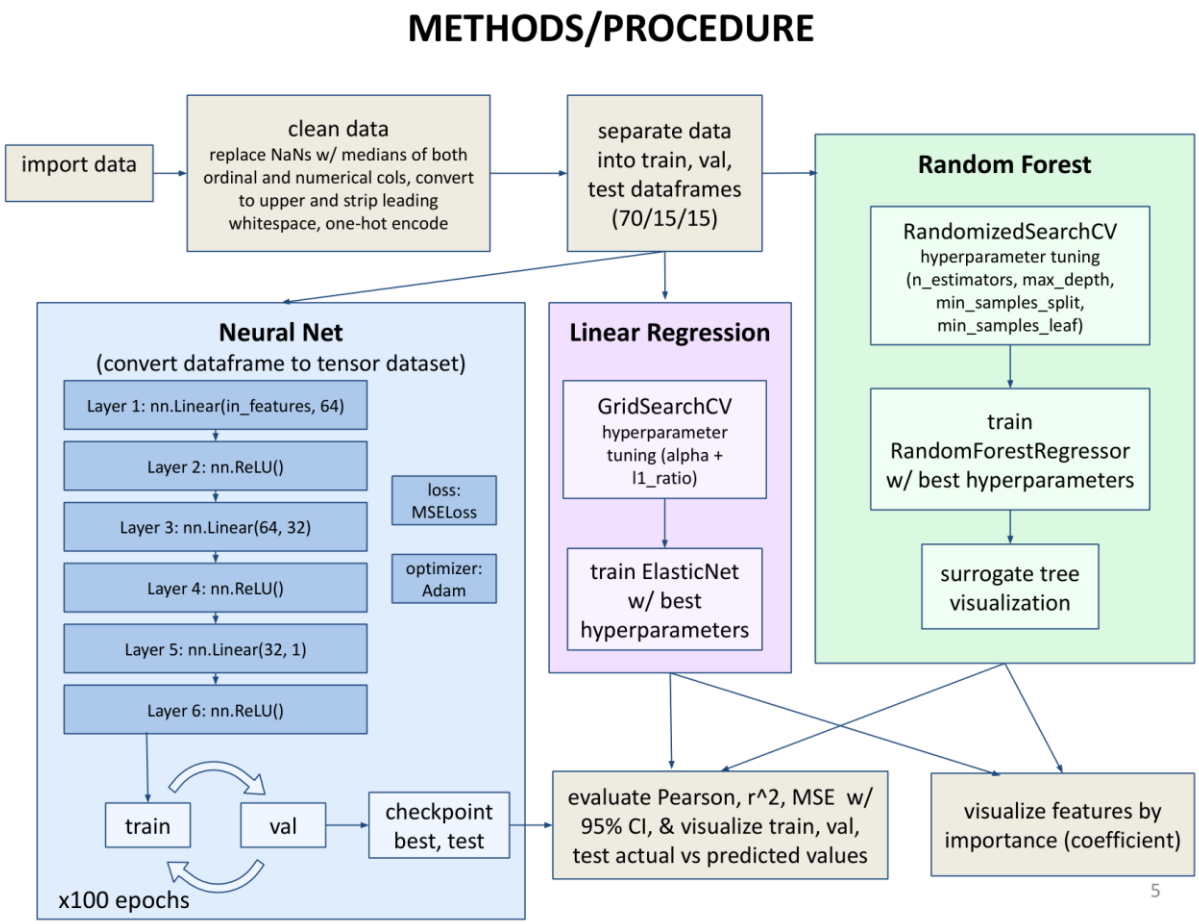

Figure 1. Methods/Procedure workflow diagram.

Step-by-step procedure:

1. Obtained METABRIC data from Kaggle and saved the raw CSV file; clean data columns to focus on gene expression features.
2. Created a patient ID mapping dictionary for row identification, then dropped patient_id from model inputs.
3. Defined target variable: time_until_death_years = overall_survival_months / 12.
4. Assessed and imputed missing values (NaNs) using median for numerical columns and mode for categorical columns.
5. Dropped outcome-related and non-useful columns to prevent data leakage: overall_survival, overall_survival_months, cohort, death_from_cancer columns, nottingham_prognostic_index.

6. Split data into train/validation/test sets (70/15/15) with random seed 42; test set reserved for final evaluation only.
7. Evaluated feature-target correlations to guide model selection.
8. Trained three models: (a) ElasticNet linear regression with GridSearchCV hyperparameter tuning (alpha, l1_ratio); (b) Random forest regression with RandomizedSearchCV tuning (n_estimators, max_depth, min_samples_split, min_samples_leaf); (c) PyTorch neural network with 6 layers, ReLU activation, MSE loss, Adam optimizer, 100 epochs with checkpointing.
9. Compared all models using Pearson correlation (r), R^2, and MSE with 95% confidence intervals on the held-out test set.
10. Extracted feature importance rankings from each model and generated surrogate decision tree visualization to interpret the Random Forest model.

Results

- Linear Regression identifies gene biomarkers responding linearly to survival duration (e.g. lymph_nodes_examined_positive
- Random Forest identifies gene biomarkers with non-linear relationships to survival duration (e.g. MAPT, CHEK1, AURKA as
- Neural Network did not perform as well as the other two models.

Model performance on held-out test set (95% confidence intervals):

Metric	Linear Regression	Random Forest	Neural Network
Test Pearson r	0.379 [0.262, 0.492]	0.392 [0.258, 0.516]	0.288 [0.148, 0.416]
Test R^2	0.138 [0.043, 0.227]	0.147 [0.039, 0.246]	0.052 [-0.061, 0.161]
Test MSE	26.187 [21.222, 31.540]	25.910 [20.815, 31.362]	37.414 [29.191, 46.972]

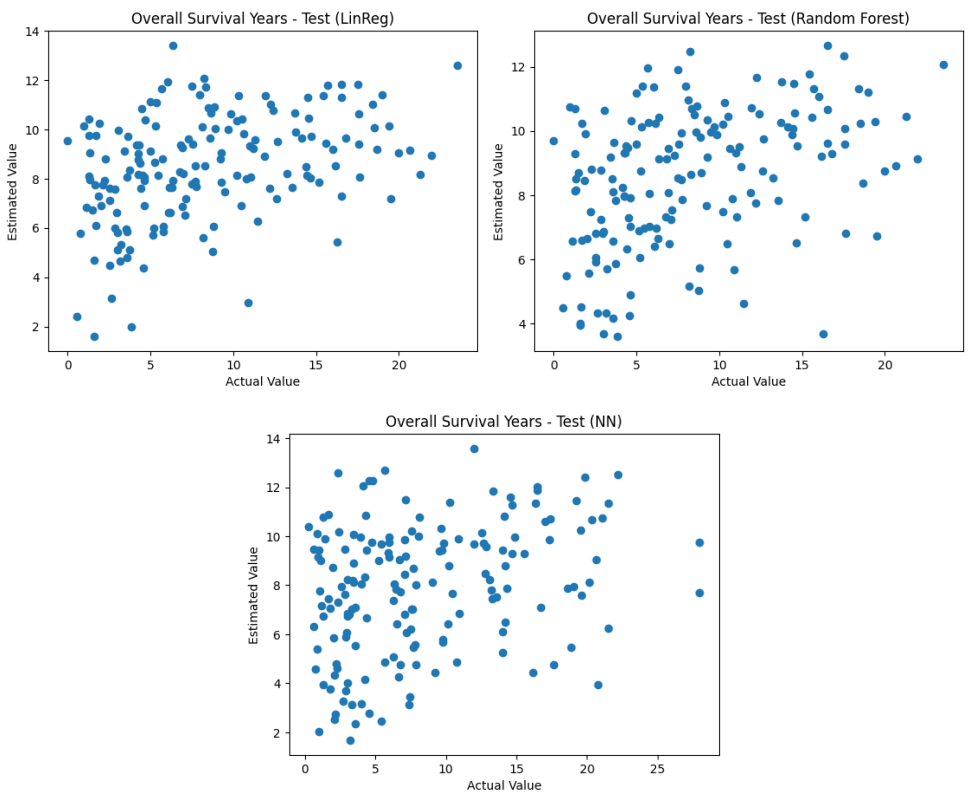

Figure 2. Actual vs. predicted survival duration for three models.

Analysis

Significant Gene Biomarkers Identified: MAPT, CHEK1, AURKA

MAPT (Microtubule-Associated Protein Tau): Encodes the Tau protein that stabilizes cellular microtubules. High MAPT expression is associated with better prognosis and improved overall survival in breast cancer (Bonneau et al., 2015). Ranked #1 in Random Forest importance, confirming its strong non-linear association with survival duration.

CHEK1 (Checkpoint Kinase 1): Encodes a serine/threonine kinase critical for DNA damage response and cell cycle regulation. Overexpression is associated with aggressive breast cancer and poor prognosis (Wu et al., 2019). Appears as root node in the surrogate decision tree, indicating it is the primary splitting factor for survival prediction.

AURKA (Aurora Kinase A): Encodes a mitotic kinase essential for cell division. High AURKA expression is associated with aggressive tumor features and shorter survival (Ali et al., 2012). Ranked in top 3 for both models.

Note: lymph_nodes_examined_positive is a traditional clinical marker showing the strongest linear relationship with survival, but it represents a conventional prognostic factor insufficient for personalized treatment. The gene biomarkers MAPT, CHEK1, and AURKA provide deeper genomic insight enabling truly personalized treatment.

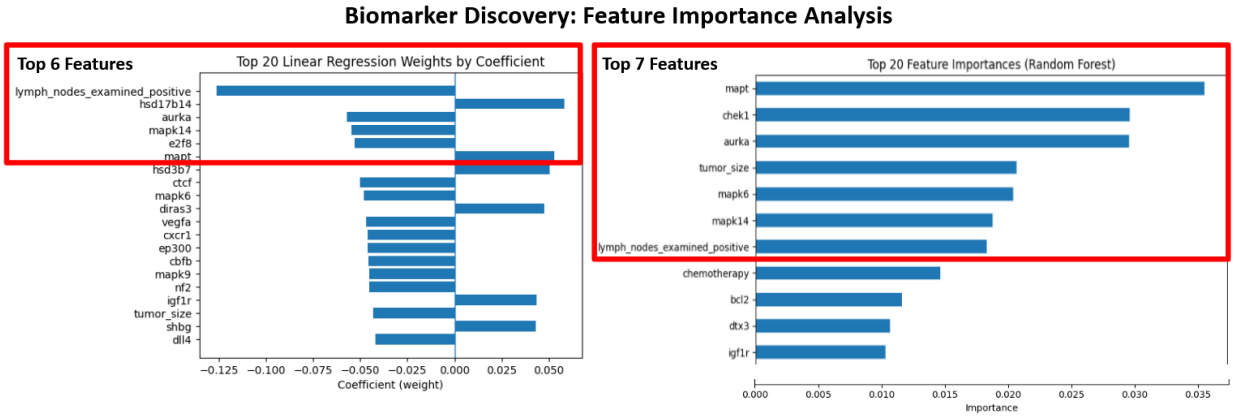

Figure 3. Feature importance analysis (Linear Regression coefficients and Random Forest importances).

Surrogate visualization

What is a Surrogate Visualization? A surrogate tree is a single interpretable decision tree trained to approximate the predictions of the Random Forest ensemble model. Because a random forest consists of hundreds of individual trees that collectively make predictions, it operates as a "black box." The surrogate tree distills the forest's complex decision-making process into a human-readable form, revealing which genes the model considers most important and how it uses them to stratify patients by predicted survival duration.

Key Observations: The root node splits on CHEK1 (threshold ≤ 0.513), confirming it as the single most important gene for initial patient stratification. Patients with lower CHEK1 expression have a mean predicted survival of 9.389 years (n=127), while those with higher CHEK1 expression have only 5.819 years (n=39). Subsequent splits involve MAP2K1, GATA3, KMT2C, and CCNE1, revealing a multi-gene decision pathway that validates the Random Forest feature importance rankings and provides clinicians with an interpretable framework for understanding gene-survival relationships.

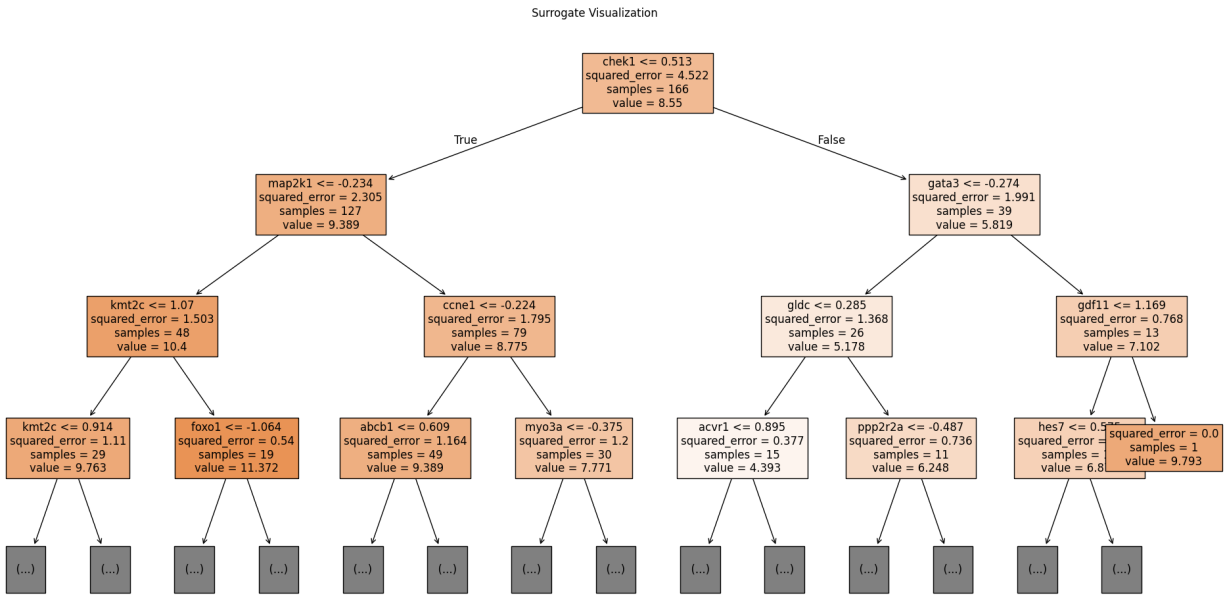

Figure 4. Surrogate decision tree approximating the Random Forest model.

Discussion

Interpretation of Results: The Neural Network did not perform as well for this small dataset (n<2,000) as anticipated, but the architecture is prepared for future improvement with much larger datasets. The multi-model approach also successfully identified three clinically validated gene biomarkers—MAPT, CHEK1, and AURKA—that correlate with breast cancer survival duration. The convergence of findings across different model types strengthens confidence in these biomarkers: MAPT and AURKA appeared in the top features of both linear regression and random forest, while CHEK1 emerged as the primary decision factor in the surrogate tree visualization.

Comparison with Published Research: These findings resonate with existing cancer genomics literature. MAPT (Tau protein) expression has been positively correlated with breast cancer survival (Callari et al., 2023; Bonneau et al., 2015). CHEK1 overexpression has been established as an independent risk biomarker associated with aggressive breast cancer phenotype (Wu et al., 2019). AURKA has been shown to outperform Ki67 as a prognostic marker in ER-positive breast cancer (Ali et al., 2012). The alignment between our computational findings and published clinical research validates the multi-model approach.

Sources of Variance and Limitations: The analysis focused primarily on genomic and gene-expression factors. However, actual survival duration depends on many additional variables including patient age, comorbidities, lifestyle factors (diet, exercise, emotional health), treatment regimens, and even unrelated causes of death (accidents, other diseases). These non-genomic factors introduce substantial variance into survival prediction. Notably, the models generally predicted survival durations lower than actual values. This under-prediction is consistent with the absence of protective factors in the model—while genomic markers capture risk, real-world outcomes are extended by factors such as effective treatment, healthcare access, and lifestyle variables that were not included. This observation highlights opportunities for future work incorporating broader clinical and demographic data.

Neural Network and Data Size Limitation: The neural network underperformed (r^2=0.052) compared to linear regression and random forest, which was anticipated given the METABRIC dataset contains fewer than 2,000 records. Neural networks require substantially larger datasets to learn complex patterns without overfitting. The training/validation loss curves confirmed overfitting, with training loss approaching zero while validation loss plateaued. For future improvement, the neural network architecture is prepared to leverage much larger genomic datasets (e.g., TCGA with 10,000+ samples) where deep learning could discover additional differential factors among larger populations.

Conclusion

Summary of Findings: This project demonstrated that a multi-model machine learning approach can successfully identify clinically significant gene biomarkers associated with breast cancer survival duration. Three key biomarkers were identified: MAPT, CHEK1, and AURKA, all of which have been independently validated in published cancer genomics research.

Addressing the Hypothesis: The hypothesis was partially supported. As predicted, different models contributed complementary findings: linear regression identified genes with proportional effects on survival, while random forest revealed non-linear gene interactions. The neural network did not perform well due to the small dataset size, as anticipated, but remains prepared for future application with larger datasets.

Clinical Significance and Applications: These biomarkers could enable physicians to personalize breast cancer treatment by assessing individual patients' MAPT, CHEK1, and AURKA expression levels. For example, patients with high CHEK1 expression (associated with poor prognosis) might benefit from more aggressive treatment or CHEK1 inhibitor therapy, while patients with high MAPT expression (associated with better prognosis) might be candidates for less intensive treatment regimens. This personalized approach could improve survival duration and quality of life for breast cancer patients.

Program Comparison: Unlike existing programs that primarily predict binary survival outcomes (alive vs. deceased), this multi-model approach predicts continuous survival duration and identifies specific gene biomarkers. This provides more actionable clinical information for treatment personalization.

Recommendations

Future research will incorporate larger datasets (e.g., TCGA), additional clinical variables, and survival-analysis models that account for censoring to further improve prediction accuracy and biomarker discovery.

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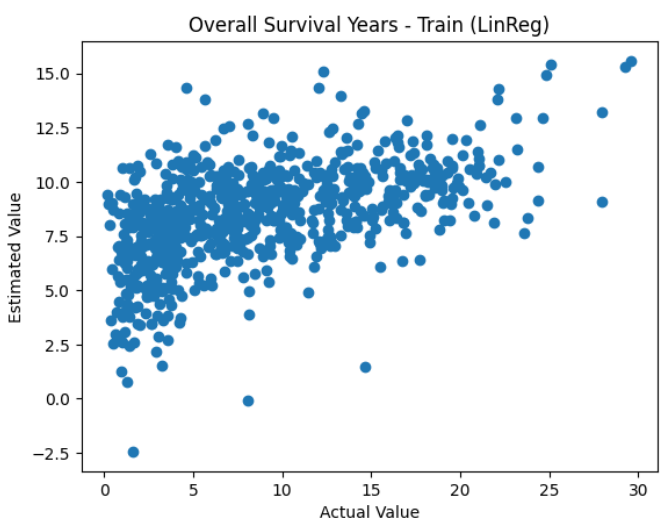
Appendices

linear regression and random forest models:
https://colab.research.google.com/drive/1j0h2OSfwsXhAsSI-JdgWvibvioufO8CF?usp=sharing
neural net model:
https://drive.google.com/file/d/1FonjdpIVNSRKlz5okLzQgVr9u8kdKi8o/view?usp=sharing

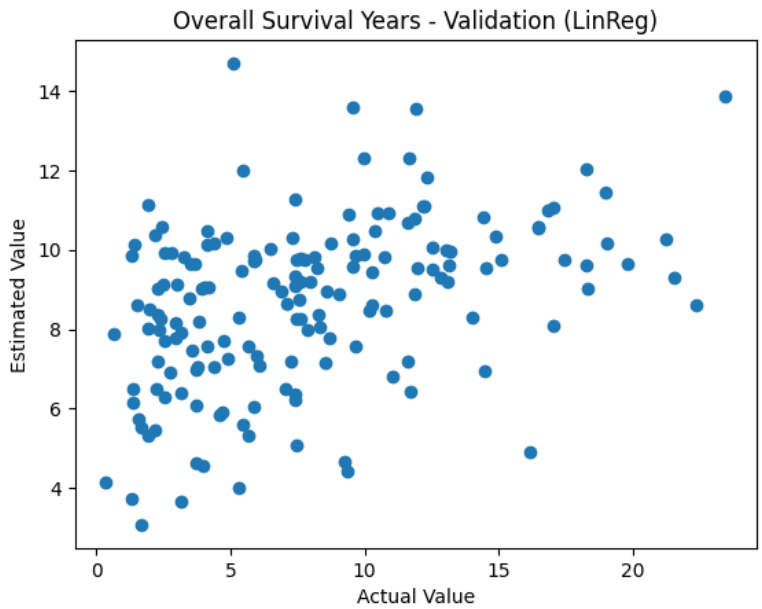
Linear Regression Results

Best parameters: {'alpha': 10, 'l1_ratio': 0.0}

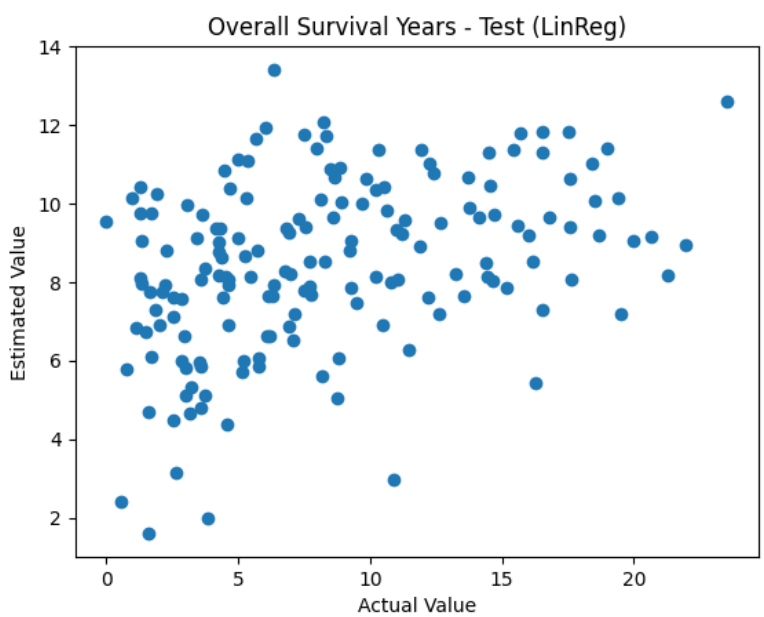
Train (LinReg) set:
pearson = 0.5510480168056319
r2 = 0.2747331555497594
mse = 25.83242349537969



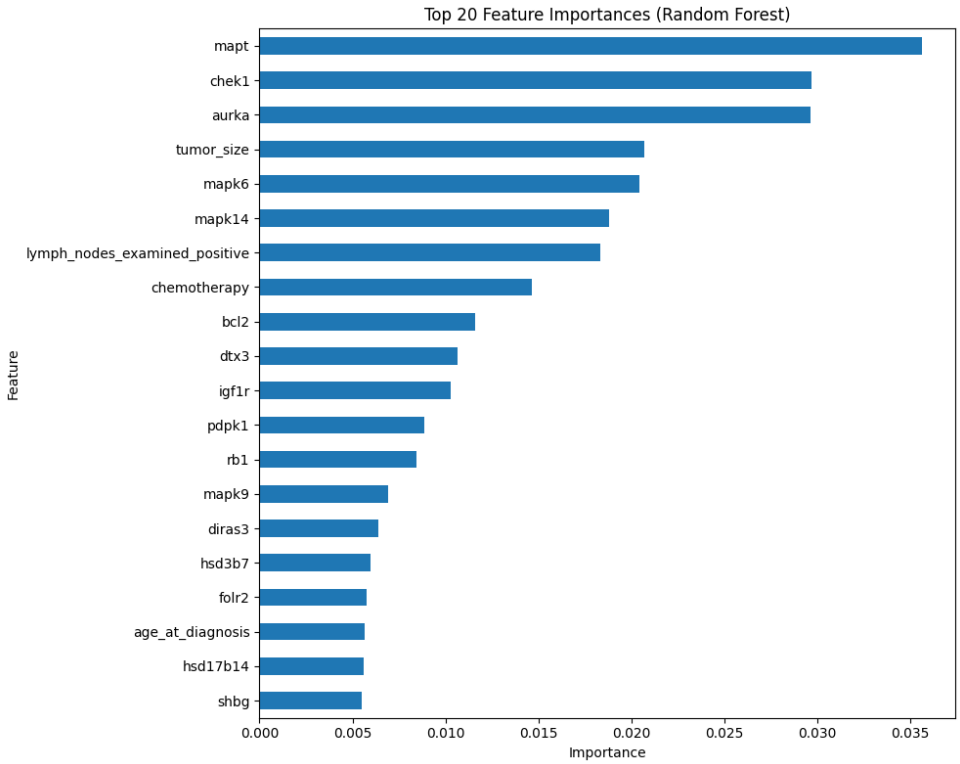
Validation (LinReg) set:
pearson = 0.41311923457762173
r2 = 0.15051870125983857
mse = 22.90454017519344

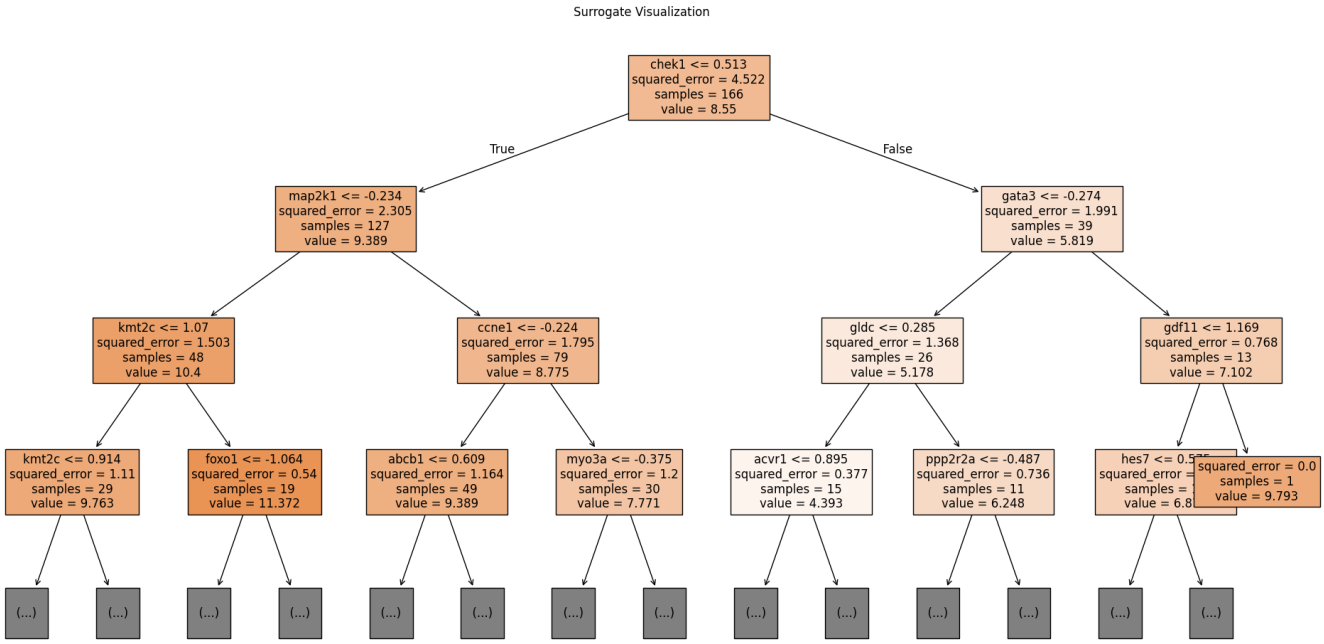


Test (LinReg) set:
pearson = 0.3807195722895215
r2 = 0.14454712152077942
mse = 26.230475351475455

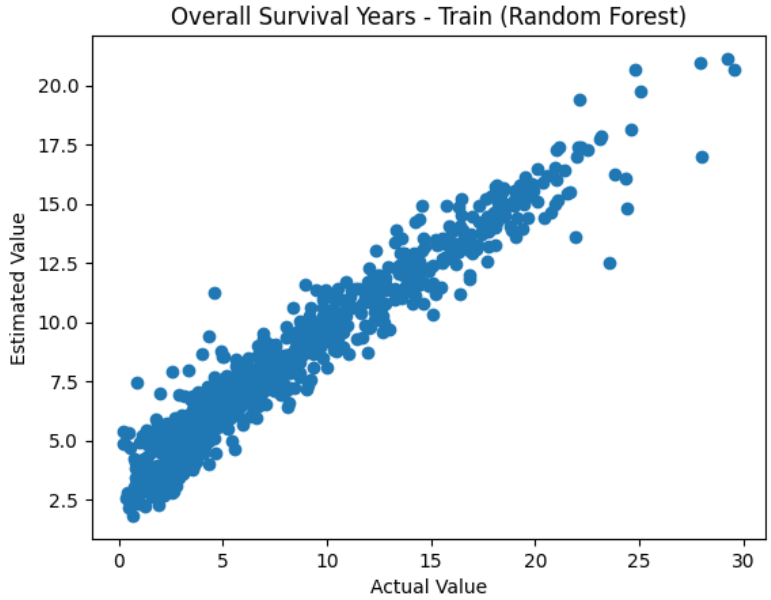


Random Forest Results

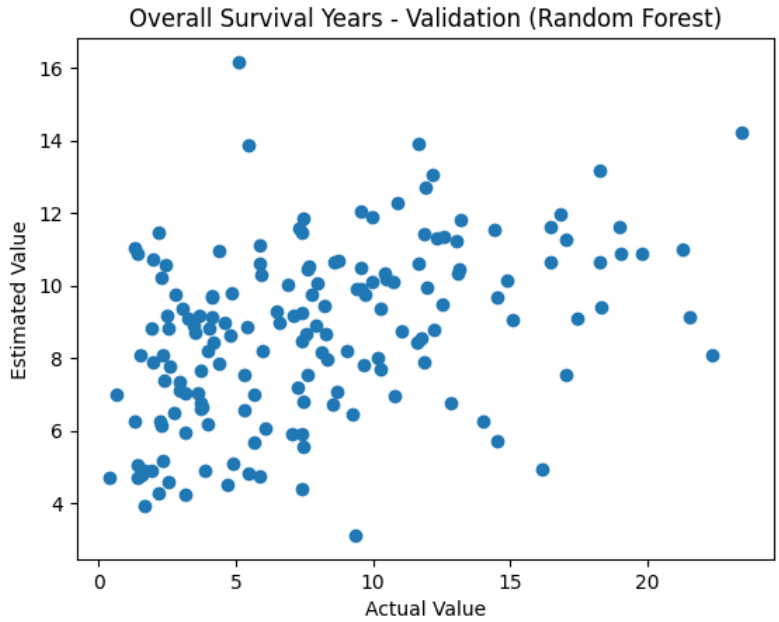




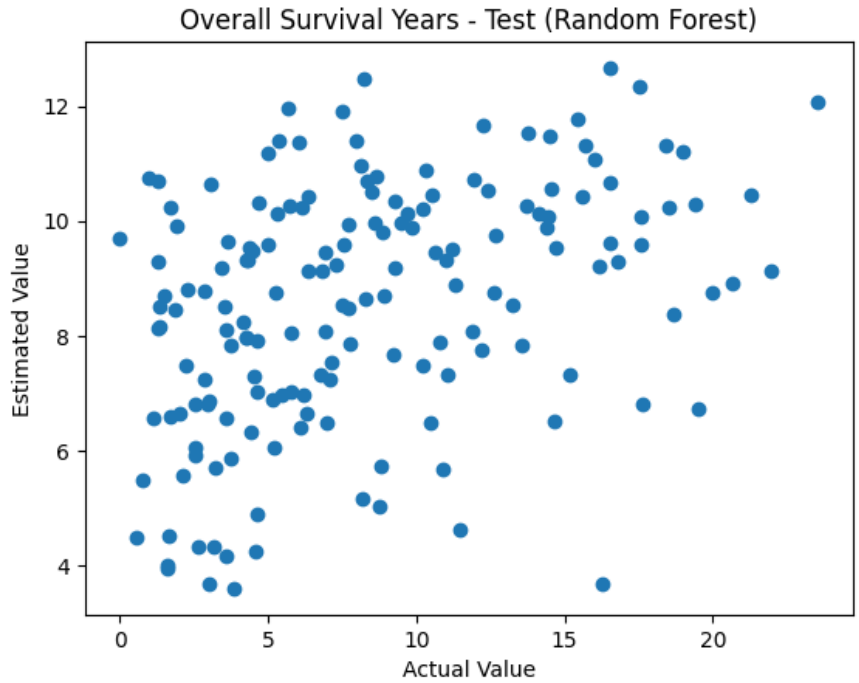
Train (Random Forest) set:
pearson = 0.9654569514802639
r2 = 0.8272532366259353
mse = 6.152863022874812



Validation (Random Forest) set:
pearson = 0.41999188920633224
r2 = 0.1537498515876109
mse = 22.81741876050861

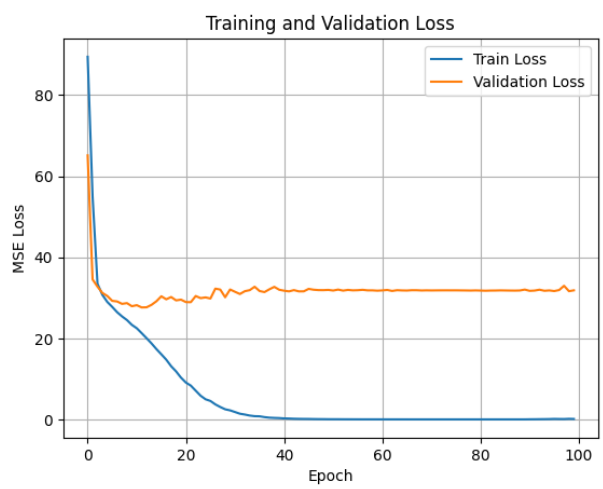


Test (Random Forest) set:
pearson = 0.392715961032233
r2 = 0.15303161661430453
mse = 25.97031801841923



Neural Net Results

Epoch 1/100 - Train Loss: 89.4710, Val Loss: 65.1409
Epoch 10/100 - Train Loss: 23.3237, Val Loss: 27.9235
Epoch 20/100 - Train Loss: 10.3030, Val Loss: 29.5438
Epoch 30/100 - Train Loss: 2.2201, Val Loss: 32.0502
Epoch 40/100 - Train Loss: 0.3200, Val Loss: 31.9879
Epoch 50/100 - Train Loss: 0.0407, Val Loss: 31.9455
Epoch 60/100 - Train Loss: 0.0095, Val Loss: 31.7576
Epoch 70/100 - Train Loss: 0.0015, Val Loss: 31.8377
Epoch 80/100 - Train Loss: 0.0005, Val Loss: 31.8215
Epoch 90/100 - Train Loss: 0.0011, Val Loss: 32.0260
Epoch 100/100 - Train Loss: 0.1047, Val Loss: 31.8436



Train (NN) set:
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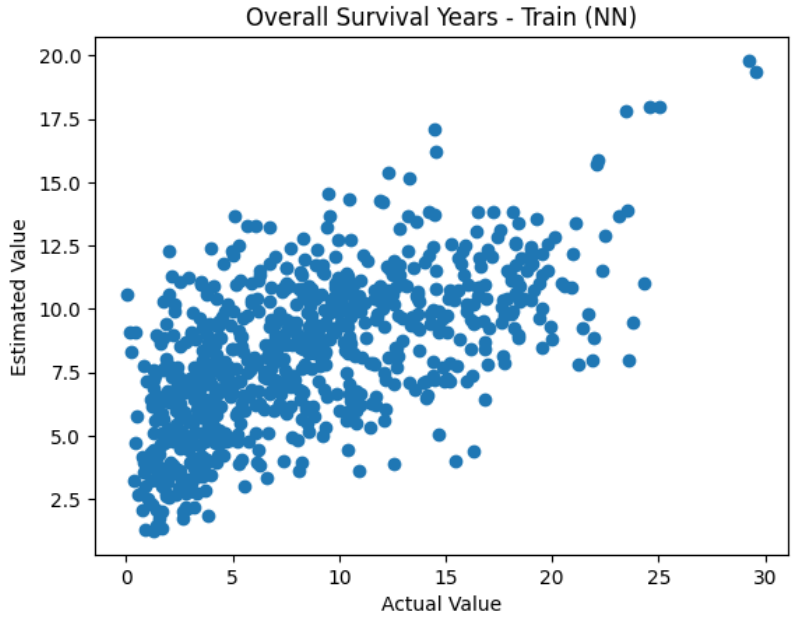
r2 = 0.3751286268234253
mse = 20.33391761779785



Validation (NN) set:
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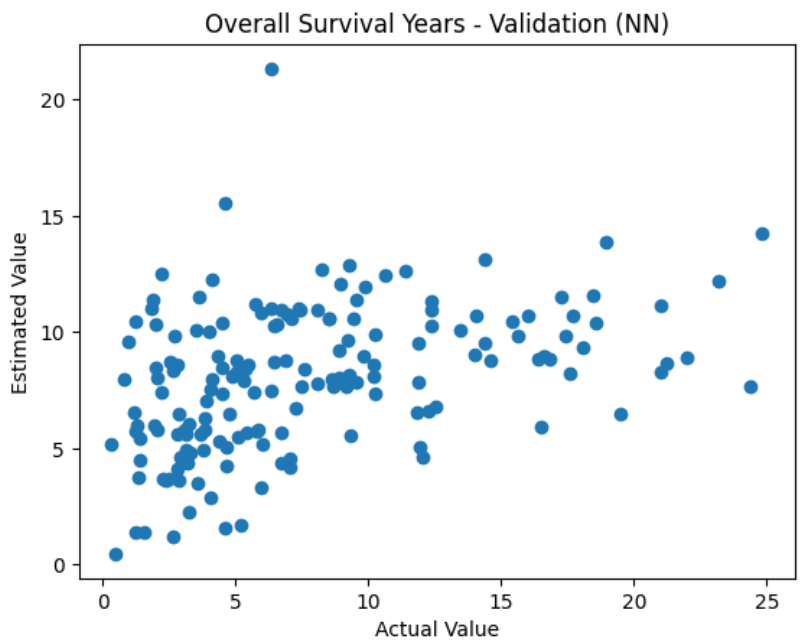
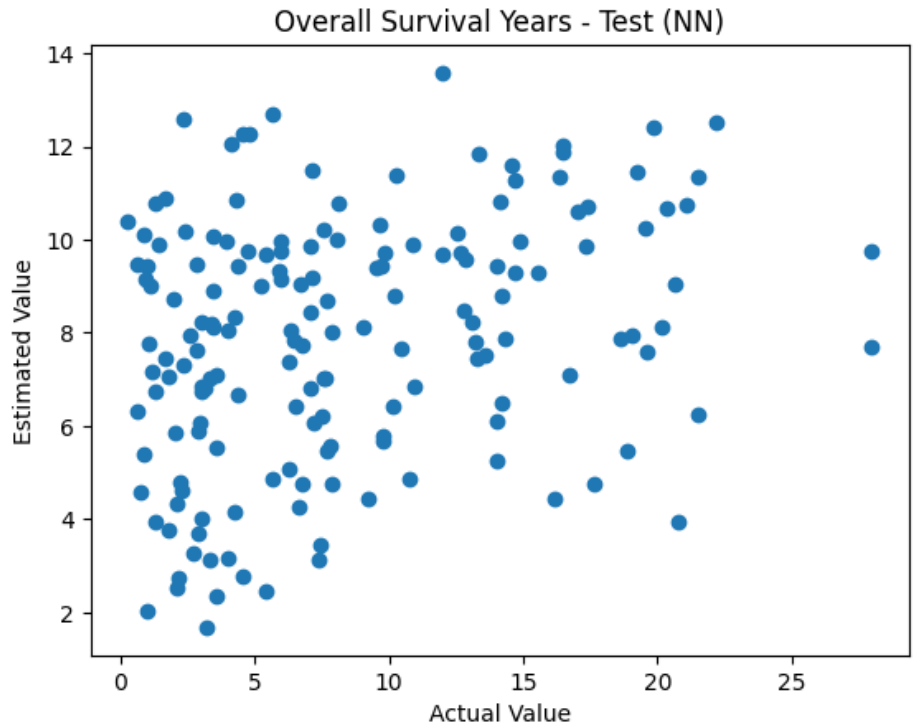
r2 = 0.13945919275283813
mse = 27.62317657470703



Test (NN) set:
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r2 = 0.056706368923187256
mse = 37.48176956176758